

Generation Interconnection System Impact Study Report

For Queue Project Q11-0042

Shroud 100 kV
November 22, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0042. The Interconnection Customer proposes a 209.5 MW wind facility interconnecting at the Shroud 100 kV substation in the Indiana area. The request was received on May 27, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 34 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,759,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ELM CRK-NACODOCHES ROAD 100.0 kV	SHROUD-SOUTHEAST CHICAGO ENERGY PROJECT 100.0 kV Line	176.3	123	51.9
SOUTHEAST CHICAGO ENERGY PROJECT-SHROUD 100.0 kV Line	SHROUD-ELM CREEK 100.0 kV	161.5	182	65.1
ELM CREEK-SHROUD 100.0 kV	SHROUD-SOUTHEAST CHICAGO ENERGY PROJECT 100.0 kV	147.5	177	51.9
HUISKAMP-HEGEWISCH SUBSTATION 100.0 kV	SOUTHEAST CHICAGO ENERGY PROJECT SUBSTATION-SHROUD 100.0 kV	126.1	137	28.1
WEST MEMPHIS GATEWAY SUBSTATION-HUGO SUBSTATION 100.0 kV	FISK ST-WESTON VILLAGE 230.0 kV L	116.4	156	16.1
TOWER AUTOMOTIVE-HEGEWISCH 100.0 kV	HEGEWISCH SUBSTATION-TOWER AUTOMOTIVE 100.0 kV Line 2	115.1	193	25.3
TOWER AUTOMOTIVE-HEGEWISCH 100.0 kV Cir. 2	HEGEWISCH-TOWER AUTOMOTIVE 100.0 kV	115.1	193	25.3
LAS VEGAS 6306-NACOGDOCHES RD SUBSTATION 100.0 kV	WESTON VILLAGE-FISK STREET 230.0 kV L	113.7	142	12.9
BORDERTOWN-WATERFLOW 100.0 kV Circuit 2	WATERFLOW-BORDERTOWN 100.0 kV #1	108.2	154	8.8
WATERFLOW-WARRENSBURG 100.0 kV Line 1	WATERFLOW-WARRENSBURG 100.0 kV line	107.8	145	9.1
YABUCA-FLETCHER CHALLENGE 100.0 kV Line	WESTON VILLAGE SUBSTATION-FISK STREET 230.0 kV	106.8	134	11.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WATERFLOW-BORDERTOWN 100.0 kV Circuit 1	WATERFLOW-BORDERTOWN 100.0 kV Ckt 2	106.5	156	8.8
WATERFLOW-WARRENSBURG 100.0 kV Cir. 2	WATERFLOW-WARRENSBURG 100.0 kV Circuit 1	104.5	149	9.1
WILDWOOD-HUISKAMP 100.0 kV	SOUTHEAST CHICAGO ENERGY PROJECT-SHROUD 100.0 kV L	102.5	119	28.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ELM CRK-NACODOCHES ROAD 100.0 kV	Rebuild line with higher-capacity conductor	\$1,637,000	69% (\$1,133,000)
SOUTHEAST CHICAGO ENERGY PROJECT-SHROUD 100.0 kV Line	Rebuild line with higher-capacity conductor	\$1,341,000	69% (\$928,000)
ELM CREEK-SHROUD 100.0 kV	Rebuild line with higher-capacity conductor	\$910,000	69% (\$630,000)
HUISKAMP-HEGEWISCH 100.0 kV	Rebuild line with higher-capacity conductor	\$1,365,000	69% (\$945,000)
W MEMPHIS GATEWAY-HUGO 100.0 kV	Rebuild line with higher-capacity conductor	\$1,371,000	69% (\$949,000)
TOWER AUTOMOTIVE-HEGEWISCH 100.0 kV #1	Rebuild line with higher-capacity conductor	\$1,542,000	69% (\$1,067,000)
TOWER AUTOMOTIVE-HEGEWISCH 100.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,697,000	69% (\$1,866,000)
NACOGDOCHES RD SUBSTATION-LAS VEGAS 6306 100.0 kV	Rebuild line with higher-capacity conductor	\$2,310,000	69% (\$1,599,000)
BORDERTOWN-WATERFLOW 100.0 kV Circuit 2	Reconductor line	\$458,000	69% (\$317,000)
WATERFLOW-WARRENSBURG 100.0 kV Line 1	Reconductor line	\$345,000	69% (\$239,000)
YABUCOA-FLETCHER CHALLENGE 100.0 kV	Reconductor line	\$329,000	69% (\$227,000)
BORDERTOWN-WATERFLOW 100.0 kV Cir. 1	Reconductor line	\$357,000	69% (\$247,000)
WATERFLOW-WARRENSBURG 100.0 kV Cir. 2	Reconductor line	\$221,000	69% (\$153,000)
HUISKAMP-WILDWOOD 100.0 kV	Reconductor line	\$663,000	69% (\$459,000)
	Total allocated to project		\$10,759,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.